Program : **Diploma in Textile Technology**	
Course Code : **5063B**	Course Title: **Textile Machine Maintenance**
Semester : **5**	Credits: **4**
Course Category: **Program Elective**	
Periods per week: **4 (L:4, T:0, P:0)**	Periods per semester: **60**

Course Objectives:

- To impart the knowledge of various Maintenance concepts , Erection procedures, application of lubricants, bearing and tools in textile industry.
- To develop the idea of maintenance schedule of different textile machines, its influence on product quality and the role of humidification plant, compressor, 5S system in manufacturing process.

Course Prerequisites:

Topic	Course code	Course Title	Semester
Knowledge of mechanical elements		Basic Textile mechanics lab	2
Machines and process in yarn manufacturing		Yarn Manufacture I,II	3,4
Machines and process in fabric manufacturing		Fabric Manufacture I,II	3,4

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Explain the various types of maintenance operation, organisation structure and the role of lubricants, bearing and tools in textile industry.	14	Understanding
CO2	Demonstrate gauges & equipments related to maintenance, erection procedure of textile machines, vibration, maintenance of card clothing and top roller.	15	Applying
CO3	Discuss the maintenance schedule of Preparatory and spinning machines, its influence on quality and the role of humidification plant, compressor and safety equipments in spinning mill.	14	Applying

CO4	Demonstrate the maintenance schedule of weaving preparatory, weaving and processing machines, its influence on quality and the role of 5S.	15	Applying
	Series Test	2	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	1						
CO2	3						
CO3		3					
CO4		3					

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**Explain the various types of maintenance operation, organisation structure and the role of lubricants, bearing and tools in textile industry.**		
M1.01	Understand the objectives and types of maintenance	3	Understanding
M1.02	Understand the organization structure of maintenance department and duties of maintenance personnel	3	Understanding
M1.03	Understand the function and types of lubricants.	4	Understanding
M1.04	Understand the types of bearing and its application in textile machines	4	Understanding

Contents:

Define maintenance - objectives - 2 systems of maintenance - Draw back & benefits - Types of maintenance - Routine, Preventive, Predictive maintenance, Remedial, Restorative maintenance, Emergency maintenance- Examples of different maintenance activities in spinning mill under each type - Maintenance man power norms per 1000 spindles for cotton mill - Maintenance Organization structure of 50,000 spindleage mill and weaving department with 100 looms -

General Duties of Maintenance supervisor - Define lubricants - functions - broad Classification - Properties of liquid lubricant - Define ISO VG classification of industrial

oil based on viscosity grade - Factors influencing selection of lubricating oil for gear box - Need for grease and solid lubricants - Commonly used solid lubricants - Different Oil and grease grade, viscosity in Cst, their area of application in spinning mill - Define bearing - Structure – function - Classification of bearings based on Rolling element -Types of Ball bearings and Roller bearings - Causes of bearing failure - Application of bearing in textile machines - Maintenance of chain, gear and gear box.			
CO2	**Illustrate the application of gauges & equipments, erection procedure of textile machines, vibration, maintenance of card clothing and top roller.**		
M2.01	Understand the concept of leveling, vibration, damping ,balancing and function of the levelling instruments	4	Remembering
M2.02	Understand the role of fasteners, general tools, equipments for maintenance and erection	4	Understanding
M2.03	Discuss the erection procedure of textile machines	4	Understanding
M2.04	Illustrate the maintenance of card clothing and top roller	3	Understanding
	Series Test I	1	
Contents: Purpose of leveling - Role of foundation - Construction and working of Spirit Level - Role of Plumb bob, wedge with spirit level, Tri square - Vibration - Types of vibration - Causes of vibrations - Damping and its methods - Define balancing and its types - Define Temporary fasteners - Role of each type - function of oil seal and gaskets - General Tools for maintenance and erection - Types of spanner and its available sizes, Torque wrench, Allen key, Hammers and its types, File, Circlip Pliers, Oil cans, Grease gun - General gauges for maintenance - Construction and working of Roller eccentricity tester - Function of ATIRA Nilometer, TARP gauge, Tachometer, Feeler gauge, Depth gauge, Plug gauge, Vernier calliper, Micrometer. General erection equipments - General points of erection - Arranging components & Work table - Provisions for power and pneumatic lines - Floor leveling - Machine layout line marking - Basic Erection sequence - Positioning the base machine - leveling & settings - Commissioning - Trial run - Basic erection procedure - carding machine, Ring frame, Power loom - Card Wire mounting, Stripping and Grinding machines -Top Roller Maintenance - Cots mounting - buffing - Berkolisation - Testing of Top Roller Concentricity, Shore Hardness and Surface Roughness.			
CO3	**Discuss the maintenance schedule of Preparatory and spinning machines, its influence on quality and the role of humidification plant, compressor and safety equipments in spinning mill.**		
M3.01	Understand the routine maintenance schedule of machines used in spinning mill.	3	Remembering
M3.02	Illustrate the defects due to improper maintenance of spinning machines	4	Understanding

<table>
<tr><td>M3.03</td><td>Discuss the role of humidification plant and compressor in spinning mill</td><td>4</td><td>Understanding</td></tr>
<tr><td>M3.04</td><td>Summarize the causes of accidents in textile mills and the role of safety equipments</td><td>3</td><td>Understanding</td></tr>
<tr><td colspan="4">Contents:
Routine maintenance schedule of Blow room, Carding, Drawing, Combing, Speed frame, Ring frame and Auto coner - Defects due to improper maintenance - Low blow room cleaning efficiency, low nep removal efficiency in card, more sliver breakages in card, high drawing sliver unevenness, roller lapping in draw frame, more roving end breakage, high roving hank CV %, Ratching in speed frame, More Yarn end breakage in Ring spinning - Need for controlling temperature and humidity - Recommended R H in different departments of cotton mill- Lay out of humidification plant - Function of each parts- Damper, supply air fan, water pump, water spray humidifier (air washer) eliminator, duct, diffuser, return air fan - Flowchart of compressed air system in mill - objectives of compressor and its specification (air pressure , compressor capacity) - Air receiver, dryer, Pressure drop and its control.
Causes of accidents in textile mills - Protective equipment for head, eye, face, ear, hands, foot and respiratory - general safety - fire alarms, extinguishers</td></tr>
<tr><td>CO4</td><td colspan="3">Illustrate the maintenance schedule of weaving preparatory, weaving and processing machines, its influence on quality and the role of 5S.</td></tr>
<tr><td>M4.01</td><td>Understand the routine maintenance schedule of weaving preparatory and weaving machine.</td><td>3</td><td>Remembering</td></tr>
<tr><td>M4.02</td><td>Illustrate the defects due to improper maintenance.</td><td>4</td><td>Understanding</td></tr>
<tr><td>M4.03</td><td>Understand the maintenance schedule of processing machines</td><td>4</td><td>Understanding</td></tr>
<tr><td>M4.04</td><td>Understand the concept of Housekeeping, 5S and its advantages in textile industry</td><td>4</td><td>Applying</td></tr>
<tr><td></td><td>Series Test II</td><td>1</td><td></td></tr>
<tr><td colspan="4">Contents:
Maintenance schedule of Pirn winding machine, Direct Warping , Sizing machine, High speed automatic shuttle Loom, Rapier and projectile loom - Care and maintenance of loom parts Heald, Reed, Picker, Shuttle, Buffer - Defects due to improper maintenance of preparatory machines and looms .
Sizing machine - more ends out of lease , excessive size droppings, Lapping on squeeze rolls and warp beams, sticky beam , Shuttle loom- machine banging, starting mark, temple cracks, double pick , weft transfer failure in rapier machine, jammed projectile
Up keep of Boilers and Kiers, Padding mangles, Stenter - Maintenance schedule of Dyeing machine - Introduction to 5S work practices - steps involved in implementation of 5S - Advantages of 5S in textile industry - Points to observed for good house keeping - stacking of heald frames, reeds, storage of size beams.</td></tr>
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Text / Reference:

T/R	Book Title/Author
T1	Neeraj Niijjaawan and Rashmi Niijjaawan - Modern approach to maintenance in Spinning
R2	Ratnam T V, Ramachandran M and Narayanaswamy G -Maintenance management in spinning – SITRA
R3	SITRA - Maintenance management in Weaving
R4	Joshi B B, Vora K B and Chetvani H G, "Spinning, Weaving, Processing machinery maintenance in Textile Mills",
R5	Textile machinery manufacturers Erection manuals and handouts

Online resources:

Sl.No	Website Link
1	www.textilemates.com
2	https://textilelearner.blogspot.com